An

Internship Report

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
OF

THIRD YEAR

IN

COMPUTER ENGINEERING

SUBMITTED BY

MAYUR UMAKANT ZOPE

UNDER THE GUIDANCE OF

Dr. VIJAY RATHI

DEPARTMENT OF COMPUTER ENGINEERING
(NBA ACCREDITED)

NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING & TECHNOLOGY

TALEGAON DABHADE TAL. MAVAL, PUNE, 410507

SAVITRIBAI PHULE PUNE UNIVERSITY

2025 -2026

PCET’s & NMVPM’s

NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING & TECHNOLOGY

TALEGAON DABHADE, PUNE – 410507

CERTIFICATE

This is to certify that, **MAYUR UMAKANT ZOPE (Roll No. TECA171)** has successfully completed the Internship entitled Frontend Development and UI Implementation using React.js under my supervision, in the partial fulfilment of the requirements of Third Year in Computer Engineering of Savitribai Phule Pune University (SPPU).

Date: / / 2026

Place: Talegaon, Pune.

Dr. Vijay Rathi	Prof. Nilesh Kambale	Dr. Rahul Patil
Internship Guide	**Internship Coordinator**	**HOD Computer Engineering**

ACKNOWLEDGEMENT

I would like to express my deepest gratitude and appreciation to all those who have contributed to the successful completion of this internship. It is with great pleasure that I acknowledge their invaluable support, guidance, and encouragement throughout this journey.

I sincerely thank my internship guide, ***Dr. Vijay Rathi***, for her exceptional support, guidance, and patience. Her expertise and mentorship have been invaluable in shaping my understanding and helping me overcome challenges.

I would also like to express my gratitude to the internship coordinator, ***Prof. Nilesh Kambale***, for his valuable suggestions, encouragement, and motivation, which contributed to the successful completion of this internship. I extend my appreciation to all faculty members for their continuous support and encouragement.

I express my heartfelt gratitude to ***Dr. Rahul Patil***, Head of the Department of Computer Engineering, for guiding me conceptually and for providing the necessary resources and constant encouragement to complete this internship work.

I am also thankful to our Principal, ***Dr. Pramod Patil***, and the Management of NMIET for providing the necessary facilities and a supportive academic environment that encouraged learning and creativity.

My sincere thanks to all the teaching and non-teaching staff of the Computer Engineering Department for their assistance and cooperation throughout this internship. I am also grateful to my friends for their support and help.

I sincerely acknowledge and appreciate the contributions of everyone who directly or indirectly supported me in completing this internship.

Mayur Umakant Zope TECA171

CONTENTS

CHAPTER I: ORGANIZATIONAL STRUCTURE OF INDUSTRY

1.1 About Weularity

Weularity is a full-service digital agency dedicated to transforming business ideas into impactful digital experiences. The agency specializes in custom website development, SaaS solutions, and strategic digital marketing. With a mission to build scalable, high-performance digital products, Weularity serves clients across various industries, helping them establish a strong online presence.

Since its inception, Weularity has successfully completed over 22 projects across domains such as real estate, business, and lifestyle. The agency operates with a lean, remote-friendly team that follows modern Agile and Scrum methodologies to deliver consistent, quality-driven results.

Weularity aims to transform business ideas into impactful digital experiences by providing custom website development, SaaS solutions, and strategic digital marketing services.

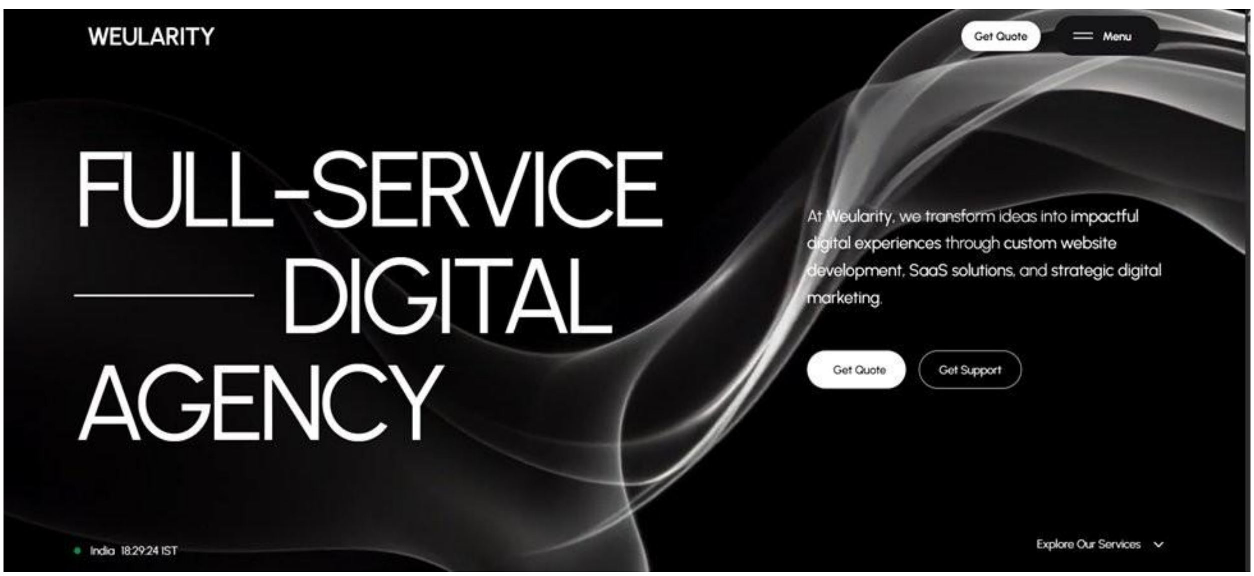

Fig 1.1: Weularity – Full-Service Digital Agency Homepage (weularity.com)

1.2 Organization Details

Organization Name	Weularity
Type	Full-Service Digital Agency
Website	weularity.com
Email	weularity@weularity.com
MSME Reg. No.	UDYAM-JK-07-0074522
Founder	Jitender Thakur
Mode of Operation	Remote
Services Offered	Web Development, UI/UX Design, Social Media & Content, Digital Marketing

Table 2.1: Organization Details

1.3 Company Vision & Mission

Weularity's vision is to transform every client's idea into a compelling digital product that drives measurable business success. The company believes that great digital experiences are not just about aesthetics — they are about performance, usability, and business impact.

The agency's mission is to deliver impactful digital experiences through custom website development, SaaS solutions, and strategic digital marketing. Weularity prides itself on marrying creativity with technology, building products that are both visually striking and functionally robust.

1.4 Services Overview

Weularity offers a comprehensive suite of digital services:

- **Web Development**: Custom frontend and full-stack solutions built with modern frameworks like React.js and Next.js.
- **UI/UX Design**: Intuitive and visually striking interface designs that enhance user experience across web and mobile platforms.
- **Social Media & Content**: Creation and management of engaging content that builds community and drives meaningful interactions.
- **Digital Marketing**: Performance-driven strategies that generate traffic, leads, and conversions for client businesses.
- **SEO Optimization:** Improving website visibility on search engines through on-page and technical SEO techniques to achieve better rankings and organic growth.
- **Branding & Identity:** Designing brand elements such as logos, color schemes, and visual identity to create a strong and consistent brand presence.
- **Business Email Solutions:** Providing professional email setup services with secure, reliable, and branded communication systems for businesses.

CHAPTER II: TECHNOLOGY LEARNED IN COMPANY

2.1 React.js

React.js is a powerful open-source JavaScript library developed and maintained by Meta (Facebook). It is used for building dynamic, component-based user interfaces. During the internship, React.js was the core frontend library used for all UI development tasks. Key concepts practiced included functional components, React hooks props and state management, component lifecycle management, conditional rendering, and event handling.

Working with React.js on production code gave firsthand experience with scalable component architecture patterns, code reusability, and maintaining consistent design systems across large codebases.

2.2 Next.js

Next.js is a production-grade React framework developed by Vercel. It extends React with powerful features such as Server-Side Rendering (SSR), Static Site Generation (SSG), and file-based routing. All projects at Weularity were built using Next.js, providing exposure to real-world application architecture. Key areas of learning included the App Router and Pages Router, dynamic routing and slug-based pages, API routes, image optimization using next/image, SEO-friendly rendering strategies, and deployment-ready configuration with next.config.mjs.

2.3 Tailwind CSS

Tailwind CSS is a utility-first CSS framework that allows rapid UI development through composable class names. Rather than writing custom CSS, Tailwind provides a vast library of utility classes that directly apply styling. During the internship, Tailwind CSS was used exclusively for all styling requirements. Concepts mastered included responsive breakpoints (sm, md, lg, xl, 2xl), flexbox and grid layout utilities, custom theme configuration via tailwind.config.js, hover and focus state management, dark mode styling, and pixel-perfect design-to-code conversion.

2.4 Git & GitHub

Version control is an essential skill in any professional development environment. At Weularity, Git and GitHub were used rigorously for all development work. The internship provided hands-on experience with creating and switching between feature branches, committing changes with meaningful commit messages, raising Pull Requests (PRs) for code review, resolving merge conflicts, and following the GitFlow branching strategy.

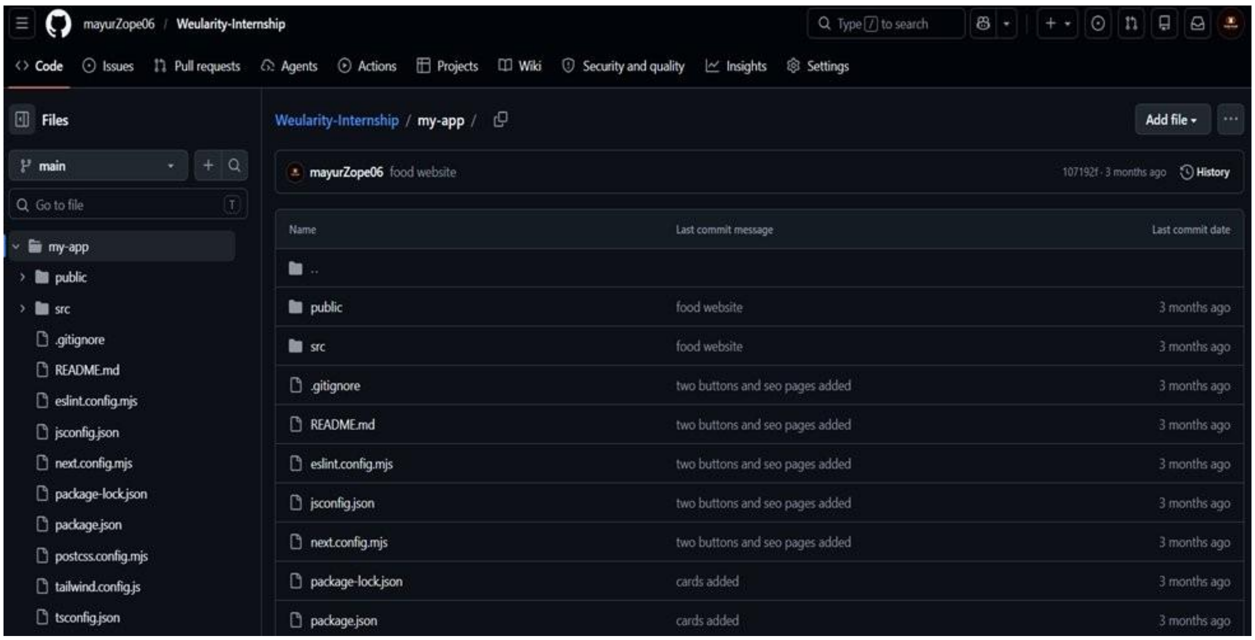

Fig 2.1: GitHub Repository – Weularity Internship Codebase (Next.js / React project structure)

The GitHub repository containing all the work completed during the internship can be accessed at: https://github.com/mayurZope06/Weularity-Internship/tree/main/my-app

This repository includes the developed React and Next.js components, responsive UI implementations using Tailwind CSS, and demonstrates the use of Git workflows such as branching, commits, and pull requests in a real-world development environment.

2.5 VS Code & Development Tools

Visual Studio Code served as the primary Integrated Development Environment (IDE) throughout the internship. Extensions like ESLint (for code quality), Prettier (for auto-formatting), GitLens (for version control insights), and Tailwind CSS IntelliSense (for class suggestions) were used to maintain code quality and development speed.

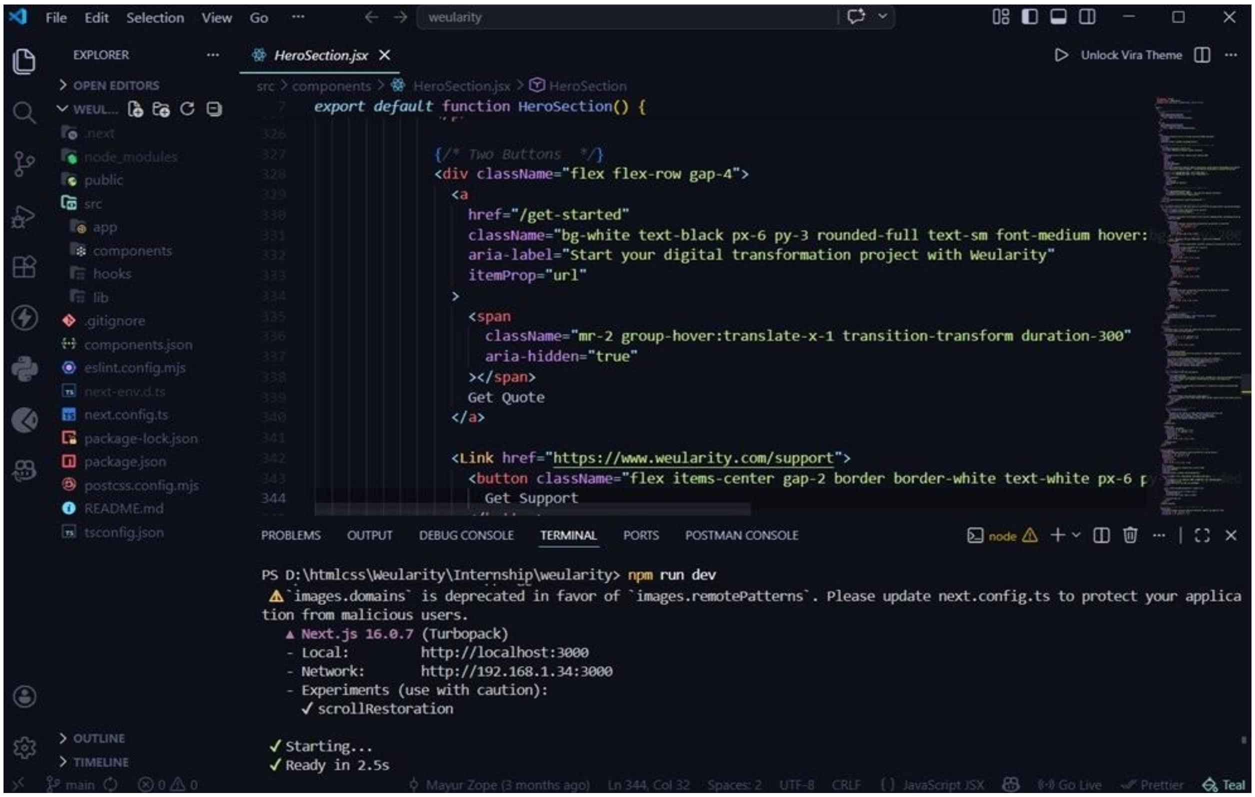

Fig 2.2: VS Code – Weularity Internship Source Code

2.6 Agile / Scrum Methodology

Weularity follows a structured Agile/Scrum workflow for project development. During the internship, I actively participated in sprint planning, regular Scrum meetings, code reviews, and sprint retrospectives, which provided practical exposure to real-world Agile practices.

I worked on assigned tasks in a sprint-based manner, maintained progress updates, and collaborated with team members through regular discussions. This experience helped me understand iterative development, the importance of continuous feedback, and effective teamwork in a professional environment.

Technology	Category	Purpose	Key Concepts
React.js	Frontend Library	Component-based UI	Hooks, State, Props
Next.js	Framework	SSR, Routing, SSG	App Router, Image Optimization
Tailwind CSS	Styling	Utility-first CSS	Responsive, Flexbox
Git & GitHub	Version Control	Code collaboration	Branching, Pull Requests (PRs)
VS Code	IDE	Development	Prettier, Source Code Structure
Agile/Scrum	Methodology	Project management	Sprints, Reviews

Table 2.2: Summary of Technologies Used During the Internship

CHAPTER III: DESCRIPTION OF INTERNSHIP

3.1 Internship Overview

The internship was undertaken at Weularity, a full-service digital agency, for the position of Frontend Developer Intern. The internship spanned one month, from 1st January 2026 to 3rd February 2026, and was conducted in a fully remote mode. It was offered as an unpaid, educational internship designed to provide practical exposure to industry-level workflows and professional development practices.

3.2 Internship Details

Intern Name	Mayur Umakant Zope
Roll Number	TEC-A171
Institution	Nutan Maharashtra Institute of Engineering and Technology, Talegaon, Pune – 410507
Degree Programme	B.E. Computer Engineering (Third Year)
Organization	Weularity – Full-Service Digital Agency
Position	Frontend Developer Intern
Duration	1st January 2026 – 3rd February 2026
Mode	Remote
Supervisor	Jai Thakur, Founder, Weularity

Table 2.2: Summary of Technologies Used During the Internship

3.3 Objective of Internship

The objective of this internship was to gain practical exposure to real-world software development, understand modern frontend technologies, and develop scalable and responsive web applications. The internship aimed to enhance technical skills in React.js, Next.js, and Tailwind CSS, while also understanding professional workflows such as Git-based collaboration and Agile/Scrum methodology.

3.4 Problem Statement

Weularity, a digital agency, required scalable and responsive frontend components for its live web platforms, weularity.com and kanect.shop. The existing codebase needed structured UI development to ensure consistency, maintainability, and improved user experience across different devices. Additionally, there was a need for accurate design-to-code conversion, where design mockups had to be implemented into functional and pixel-perfect user interfaces. The project also required proper version control practices, including structured development using Git and collaboration through pull requests, following Agile/Scrum methodology.

3.5 Scope

The scope of this internship included developing reusable frontend components, implementing responsive UI designs, converting design mockups into functional code, and contributing to live production websites. The work focused on improving user interface consistency, performance, and scalability of web applications.

3.6 Methodology

The internship followed a structured Agile/Scrum methodology with work organized into weekly sprints. Each sprint began with a planning phase where tasks were assigned, followed by development and regular progress tracking through Scrum meetings, and concluded with a review. This approach ensured consistent progress, early identification of issues, and continuous improvement in both code quality and overall development practices.

The overall workflow was:

- Sprint Planning: Tasks identified and assigned based on priority and capacity.
- Development: Writing and testing code locally, following naming conventions and component patterns established in the codebase.
- Version Control: All changes pushed to feature branches on GitHub with meaningful commit messages.
- Code Review: Pull Requests raised for review, feedback incorporated, and branches merged.
- Sprint Review: Completed work demonstrated to the team; feedback recorded for improvement.

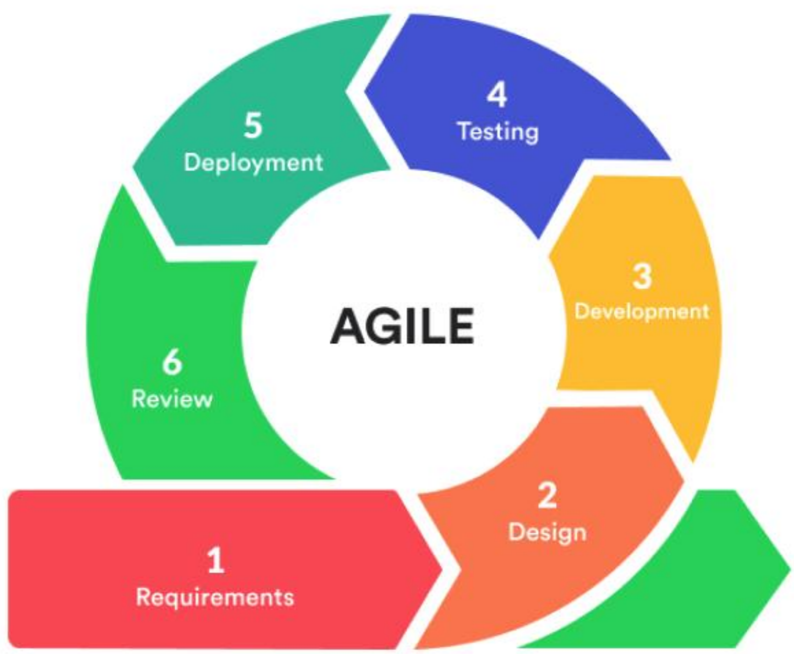

Fig 2.3: Agile Methodology

CHAPTER IV: DESCRIPTION OF TASKS

4.1 Introduction of Projects

During the internship at Weularity, contributions were made to two live production web applications:

Project 1: weularity.com – Agency Website

This is Weularity's own official company website, built with Next.js and Tailwind CSS. The website showcases the company's services, portfolio of completed projects, blog articles, and contact/quote request flows. Contributions included building UI components for the homepage hero section, the services showcase cards, the portfolio/projects gallery section, and the blog listing and individual article pages.

Project 2: kanect.shop – Business /Lifestyle Platform

Kanect.shop is a client-side project in the business and lifestyle domain built using React.js and Next.js. The project required development of a business menu/catalog system, responsive layout components for mobile and desktop views, and integration-ready UI structures for backend data.

4.2 Work Completed – Week-by-Week

Week 1 – Onboarding & Setup

- Attended virtual onboarding and orientation sessions with the Weularity team.
- Set up the development environment: Node.js, Next.js, Git, VS Code with recommended extensions.
- Cloned the existing codebase of weularity.com from GitHub and thoroughly studied the project structure, component patterns, and naming conventions.

  Existing codebase GitHub Link - https://github.com/JaiThakur10/Weularity_Startup
- Reviewed the Agile/Scrum workflow and got familiar with how sprints and PRs were managed.

Week 2 – UI Component Development

- Built dynamic, reusable React functional components with props and state management.

- Implemented file-based routing in Next.js for new pages.
- Raised the first Pull Requests and received and incorporated code review feedback.
- Developed the blog listing and article pages for weularity.com.

Fig 4.1: Blog article page built for Weularity – 'Your Website Is Your First Impression' (January 2026)

Fig 4.2: Blog article page – 'Why Thousands of Instagram Followers Still Don't Guarantee Sales' (January 2026)

Week 3 – Responsive UI & Dynamic Component Implementation

- Applied Tailwind CSS mobile-first responsive breakpoints (sm, md, lg, xl) across all components.

- Ensured all newly built components rendered seamlessly on mobile, tablet, and desktop screens.
- Tested layouts against design references side-by-side for pixel accuracy.
- Developed dynamic UI elements such as About section, project buttons, and works/portfolio sections.
- Implemented reusable React components with dynamic rendering using props and state.
- Used conditional rendering to display content dynamically based on data.

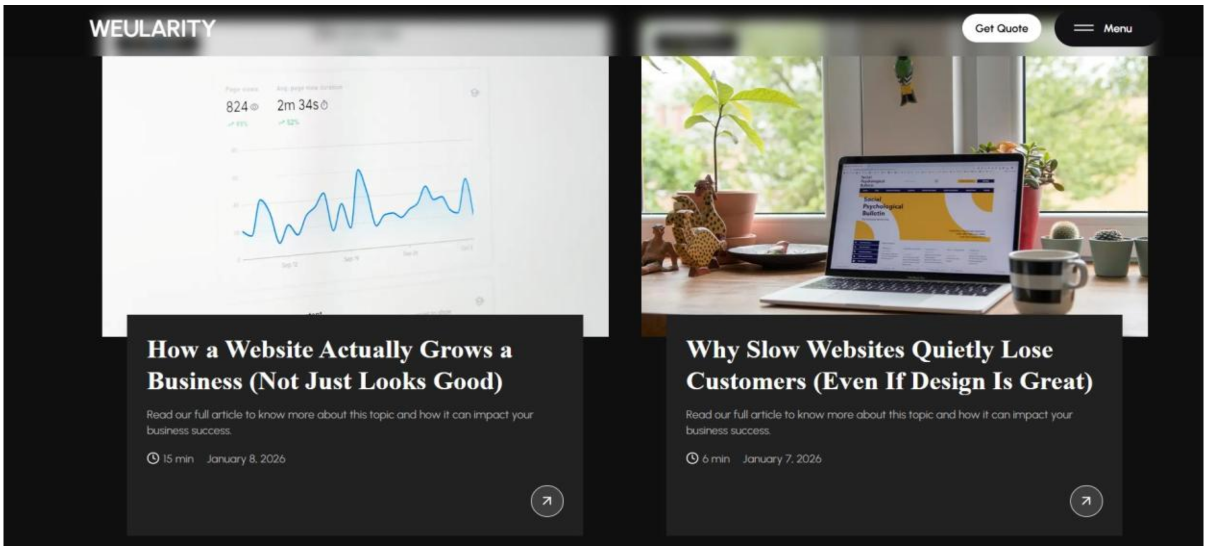

Fig 4.3: Blog listing page showing multiple articles with reading time and date indicators

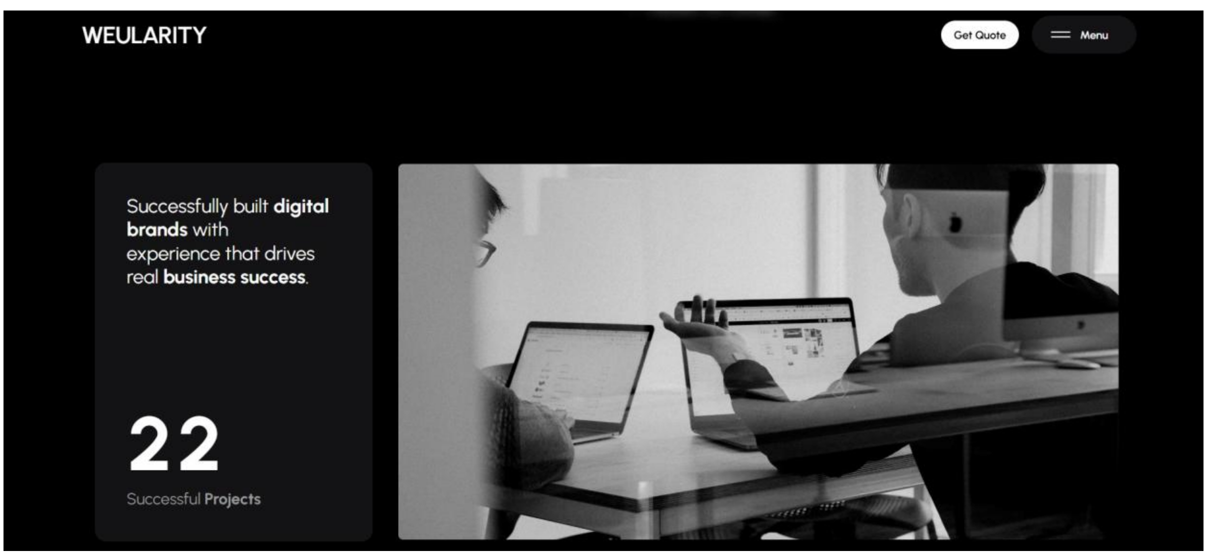

Fig 4.4: Company stats section – 'Dynamic Successful Projects' showcase on weularity.com

Week 4 – Design-to-Code Conversion & Finalization

- Converted pixel-perfect design templates into fully functional React/Next.js components.
- Implemented the portfolio/projects gallery section on weularity.com with project cards showing tags for Web Development and UI/UX Design.
- Completed integration of the business website project for the kanect.shop platform.
- Finalized all open PRs, addressed final review comments, and ensured all branches were merged cleanly.
- Participated in the final sprint review and retrospective session.

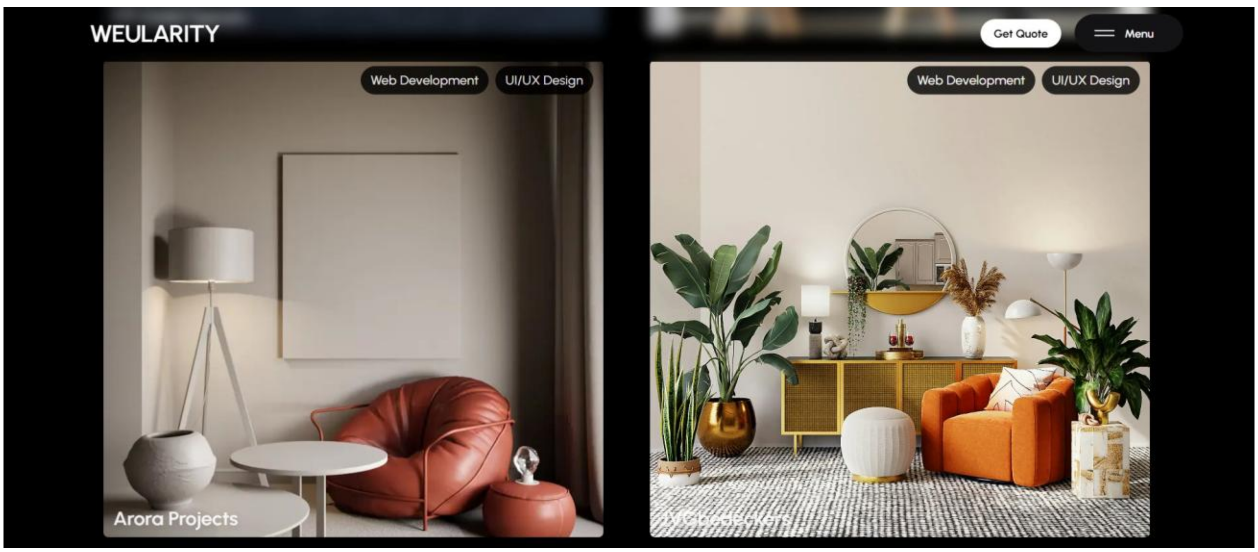

Fig 4.5: Portfolio/Projects gallery section of weularity.com – Arora Projects & Jobseekers listings

4.3 Challenges Faced & Solutions

Challenge 1 – Adapting to a Production Codebase: The existing codebase at Weularity was already structured with established patterns, component hierarchies, and naming conventions. As a new intern, understanding this codebase before making changes was critical. Solution: The entire first week was dedicated to reading existing code, studying component patterns, and understanding the naming conventions before writing any new code.

Challenge 2 – Pixel-Perfect Design Conversion: Converting design mockups exactly into code while maintaining responsiveness was a significant challenge. Solution: Tailwind CSS's fine-grained utility classes were used to match design specifications precisely, and side-by-side comparison with design references was performed at each stage.

Challenge 3 – Git Workflow in a Team Environment: Managing branches, avoiding merge conflicts, and following PR-based review processes was new territory. Solution: Following GitFlow conventions strictly, keeping commits focused and atomic, and communicating proactively with the team on branch dependencies.

4.4 Result

The work completed during the internship contributed directly to live production websites. The developed components improved UI consistency, responsiveness, and overall user experience. The contribution of multiple pull requests ensured structured and maintainable code integration into the main project. The developed components are currently part of live production websites and actively used in real-world scenarios.

CHAPTER V: LEARNING OUTCOMES

5.1 Technical Skills Developed

- **React.js Component Development:** Gained strong hands-on experience in building reusable, production-ready React functional components using hooks such as useState, useEffect, and useCallback. Learned to manage component lifecycle, handle props-based communication, and structure component hierarchies for scalability and maintainability in real production codebases.
- **Next.js Application Architecture:** Developed a thorough understanding of the Next.js framework, including the App Router, file-based routing, dynamic slug pages, Server-Side Rendering (SSR), Static Site Generation (SSG), image optimization using next/image, and SEO-friendly metadata configuration. Worked on Next.js applications deployed in a live production environment.
- **Responsive UI Design with Tailwind CSS:** Mastered the utility-first approach of Tailwind CSS to implement mobile-first, pixel-perfect, and fully responsive layouts across all breakpoints (sm, md, lg, xl, 2xl). Learned to configure custom themes, apply hover and focus states, and manage complex grid and flexbox layouts efficiently without writing custom CSS.
- **Version Control and Collaborative Git Workflow:** Gained hands-on proficiency in Git-based team workflows including creating and managing feature branches, writing meaningful and atomic commit messages, raising Pull Requests for peer review, incorporating code review feedback constructively, resolving merge conflicts, and following the GitFlow branching strategy used by the Weularity development team.
- **Design-to-Code Conversion:** Developed the ability to accurately translate UI/UX design mockups into fully functional, pixel-perfect React and Next.js components. This included matching spacing, typography, colour palettes, and interactive states exactly as defined in the design references, a critical skill in any professional frontend development role.
- **Reading and Extending Existing Codebases:** Learned the professional discipline of thoroughly understanding an established codebase before making changes. This involved studying component architecture, naming conventions, folder structures, and data flow

patterns in a live project, rather than building from scratch, which is the norm in real-world development environments.

- **SEO and Web Performance Fundamentals:** Gained exposure to SEO best practices through Next.js metadata configuration, structured page layouts, semantic HTML usage, and image optimization techniques. Understanding how frontend code directly impacts search engine visibility was a key practical insight gained during the internship.
- **Live Production Contribution:** Successfully contributed code to two live, public-facing websites — weularity.com (the agency's official website) and kanect.shop (a client food and lifestyle platform). All contributions passed peer code review and were merged into the main production branches, giving direct experience of delivering code that is used by real users.

5.2 Professional Skills Developed

- Gained firsthand experience working within an Agile/Scrum team framework.
- Developed communication skills through sprint planning meetings, daily standups, and PR review discussions.
- Learned to manage time effectively within sprint deadlines and to prioritize tasks based on impact.
- Built an appreciation for code quality, meaningful commit messages, and documentation.
- Developed the professional habit of writing clean, maintainable, and well-structured code.

CHAPTER VI: CONCLUSION

6.1 Summary

The one-month internship at Weularity as a Frontend Developer Intern was a highly rewarding and transformative experience. Working on two live production websites — weularity.com and kanect.shop — provided valuable real-world exposure beyond academic learning. Contributing reviewed and production-ready code to real client projects significantly enhanced my technical confidence and understanding of industry practices.

6.2 Key Takeaways

This internship reinforced that modern frontend development goes beyond basic HTML, CSS, and JavaScript. It involves component-based architecture, responsive design, performance optimization, version control, and effective team collaboration.

I gained hands-on experience with React.js, Next.js, Tailwind CSS, and Git workflows, which are essential skills in the current industry. Additionally, working in an Agile/Scrum environment helped me understand iterative development, continuous feedback, and professional communication within a team.

The experience also improved my ability to understand existing codebases, write clean and maintainable code, and incorporate feedback through code reviews — all of which are critical skills for a software developer.

6.3 Future Scope

The foundation built during this internship opens up numerous paths for continued growth. Future areas of exploration include advanced Next.js features, server actions and the React Server Components model, TypeScript integration for type-safe development, testing with Jest and React Testing Library, and full-stack development using Next.js API routes combined with database integrations.

CHAPTER VII: REFERENCES

[1] React.js Official Documentation. Available at: https://react.dev

[2] Next.js Official Documentation. Available at: https://nextjs.org/docs

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[4] Git Documentation. Available at: https://git-scm.com/doc

[5] GitHub Repository – Weularity Internship Codebase. Available at:

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[6] Weularity Official Website. Available at: https://weularity.com

[7] Agile Alliance – Scrum Framework. Available at: https://www.agilealliance.org/agile101/

[8] MDN Web Docs – JavaScript and Web APIs. Available at: https://developer.mozilla.org

ANNEXURE A: INTERNSHIP LOG BOOK

The following table records the weekly activities and tasks completed during the internship at Weularity & EduSkills:

Week	Work Done	Coordinator Sign
Week 1 (Jan 1-7)	Onboarding & orientation. Set up Next.js development environment. Studied codebase, Git workflow, and Agile/Scrum process. Reviewed component patterns and naming conventions.	
Week 2 (Jan 8-14)	Built dynamic React functional components. Implemented Next.js routing. Raised first Pull Requests on GitHub. Developed blog listing, article pages and services card components for weularity.com. Authored blog articles for content marketing.	
Week 3 (Jan 15-21)	Applied Tailwind CSS responsive breakpoints across all components. Fixed layout issues for cross-device compatibility. Pixel-perfect testing against design references. Merged reviewed branches.	
Week 4 (Jan 22-28)	Converted design templates to production-ready Next.js components. Completed business website for kanect.shop. Finalized all open PRs. Participated in sprint review and retrospective. Handed over work to team.	
Week 5 (Jan 28 – Feb 3)	Assisted in merging final frontend branches into production for Weularity and Kanect. Internship Completion: Successfully completed all assigned frontend tasks and internship requirements.	

ANNEXURE B: COMPLETION CERTIFICATE

1. The following is a copy of the Certificate of Internship issued by Weularity upon successful completion of the Frontend Developer Internship program:

Annexure B: Certificate of Internship issued by Weularity – Jitender Thakur, CEO & Founder

2. The following is a copy of the Certificate of Internship issued by EduSkills upon successful completion of the AI-ML Internship program:


शिक्षा मंत्रालय
MINISTRY OF
EDUCATION
सत्यमेव जयते

अखिल भारतीय तकनीकी शिक्षा परिषद्
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Karthik Padmanabhan Developer Ecosystem Lead MENA & India, Google	Dr. Buddha Chandrasekhar Chief Coordinating Officer (CCO) AICTE, Ministry of Education	Shubhajit Jagadev Chief Executive Officer (CEO) EduSkills

Certificate ID : 4db197a7d91109f11be4

Student ID: STU681ba784e31251746642820

INTERNSHIP GRADE
O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Annexure B: Certificate of Internship issued by EduSkills